

Diaphragm-type accumulator Type HAD

Component series 1X and 2X

RE 50150-B/12.11
Material no. R901317044

Replaces: -
English

Operating instructions



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The cover page shows an example configuration. The product supplied may therefore differ from the photo shown.

The original operating instructions were prepared in German.

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1 About this document

These instructions contain important information on the safe and appropriate transport, assembly, commissioning, maintenance, disassembly and simple troubleshooting of the HAD..1X/-2X diaphragm-type accumulator.

- Read these instructions completely, especially chapter "2 General safety instructions", before working with the HAD..1X/-2X diaphragm-type accumulator.

1.1 Related documents

The HAD-1X/-2X diaphragm-type accumulator is a machine or system component.

- Please also observe the instructions for the other machine or system components as well as the documentation provided by the machine or system manufacturer.
- Please also observe the following documentations:

Table 1: Related documents

Documentation	Contents
Data sheet 50150	Technical data, operating conditions, limits of performance, project planning information concerning HAD..1X/-2X diaphragm-type accumulators
Data sheet 07008	General product information on hydraulic products
Data sheet 07900	General information on the assembly, commissioning and maintenance of hydraulic systems
Data sheet 90220	General information on hydraulic fluids on mineral oil basis

You can find these operating instructions as well as the related documents in the media directory under
<http://www.boschrexroth.com/variou utilities/mediadirectory/>.

- Please also observe the generally applicable, legal or otherwise binding regulations of the European or national legislation and the regulations on accident prevention and environmental protection applicable in your country.

1.2 Abbreviations used

Table 2: Abbreviations

Abbreviation	Meaning
HAD..1X/-2X	Diaphragm-type accumulator, series 1X/-2X

1.3 Signs and symbols used

In these instructions, the following signs and symbols are used:

- ▶ Activity symbol: The text after this symbol describes activities that can be performed in any order.
- 1. Numbered activities: The text after this symbol describes activities that are to be performed in the order indicated.
The intended text after activities/an activity describes their/its result.
- 1st category bullet point
 - 2nd category bullet point



This pictogram is used to mark important information and remarks. The text contains useful information that you should particularly observe, since it improves the operating procedure or is required as background information!

Safety instructions

In these instructions, there are safety instructions before an instruction whenever there is a risk of personal injury or damage to property. The measures described for the prevention of dangers must be observed.
Safety instructions are set out as follows:

SIGNAL WORD	Type of risk
	Consequences ▶ Precautions

- Warning sign (warning triangle): Draws attention to the danger
- Signal word: Indicates the degree of the danger
- Type of risk: Specifies the type or source of the danger
- Consequences: Describes the consequences in case of non-compliance
- Precautions: Specifies how the hazard can be prevented

The signal words have the following meaning:

Table 3: Meaning of signal words

Signal word	Application
DANGER! 	Indicates an imminently hazardous situation which, if not avoided, will certainly result in serious injury or even death.
WARNING! 	Indicates a potentially hazardous situation, which, if not avoided, could result in serious injury or even death.
CAUTION! 	Indicates a potentially hazardous situation which, if not avoided, could result in minor or moderate injury or damage to equipment.
	If this information is disregarded, the operating procedure may be impaired.

2 General safety instructions

The HAD..1X/-2X diaphragm-type accumulator was manufactured according to the state-of-the-art. There is, however, still a risk of personal injury or damage to property if the following general safety instructions and warnings before instructions contained in these operating instructions are not observed.

- ▶ Read these instructions completely and thoroughly before working with the HAD..1X/-2X diaphragm-type accumulator.
- ▶ Keep these instructions in a location where they are accessible to all users at all times.

2.1 Intended use

Rexroth diaphragm-type accumulators are intended to be built into hydraulic drive systems for the purposes of energy storage, shock and vibration absorption, and leakage oil compensation or volume compensation.

Diaphragm-type accumulators are pressure equipments in the sense of the Pressure Equipment Directive 97/23/EC.

The HAD..1X/-2X diaphragm-type accumulator is exclusively intended for integration into a machine and/or system or for assembly with other components to form a machine and/or system.

During project planning, the basic principles of the Pressure Equipment Directive and the Machinery Directive, within the EU, as well as other local provisions, outside the EU, are to be observed.

The diaphragm-type accumulator may only be commissioned if it has been integrated into the machine/system for which it is intended and after it was confirmed that the machine/system complies with the provisions of the Machinery Directive, the Pressure Equipment Directive and other locally applicable provisions.

Imperatively observe the technical data, operating conditions and limits of performance specified in the data sheet 50150.

Repairs may only be carried out by the manufacturer or their authorized dealers and agencies. If this condition is not complied with, the warranty expires.

The HAD..1X/-2X diaphragm-type accumulator is a technical equipment and not intended for private use.

The intended use also includes that you have completely read and understood these instructions, in particular chapter "2 General safety instructions".

2.2 Improper use

Any use of the HAD..1X/-2X diaphragm-type accumulator other than described in chapter "2.1 Intended use" is considered as improper.

Modifications of the diaphragm-type accumulator of whatever nature are not permitted, otherwise the warranty will expire! This is particularly true for welding and soldering works, mechanical processing or the installation or addition of non-original spare parts or other accessory parts.

The operating pressure of the diaphragm-type accumulator must not exceed the maximum operating pressure indicated in the data sheet 50150.

The diaphragm-type accumulator must not be charged with a gas other than nitrogen 99.99 vol. %.

The diaphragm-type accumulator must not be operated with group 1 hydraulic fluids (explosive, inflammable, fire-accelerating, toxic) or with corrosive fluids.

General safety instructions

The diaphragm-type accumulator must not be used in hydraulic, passive braking systems (trailer brake).

The diaphragm-type must not be used in explosive environments.

2.3 Qualification of personnel

Assembly, commissioning, disassembly and maintenance (incl. servicing and care) require a basic knowledge of mechanics and hydraulics, as well as familiarity with the associated technical terms. In order to ensure operational safety, these activities may only be carried out by corresponding experts or an instructed person under the direction and supervision of an expert.

Experts are those who can recognize potential dangers and apply the appropriate safety measures due to their technical training, knowledge and experience, as well as their understanding of the relevant provisions pertaining to the work to be undertaken. An expert must observe the relevant specific professional rules.

2.4 Adhere to the following instructions

General instructions

- Observe the regulations on accident prevention and environmental protection valid in your country.
- Only use Bosch Rexroth products when they are in a technically perfect condition.
- Check the product for obvious transport damages and/or other damages.
- Do not modify the HAD..1X/-2X diaphragm-type accumulator and do not reconstruct and/or attach it in any other way.
- Only use the HAD..1X/-2X diaphragm-type accumulator in the performance range which is specified in the technical data sheet 50150 and on the embossing found on the housing.
- Persons who assemble, operate, dismantle or maintain products supplied by Bosch Rexroth must not consume any alcohol, drugs or medications that may have a detrimental effect on their ability to react.
- The HAD..1X/-2X diaphragm-type accumulator may considerably heat up during operation. Allow the diaphragm-type accumulator to cool down sufficiently before touching it. If this is not possible, protect yourself by means of heat-resistant protective clothing, e.g. gloves.
- The warranty only applies to the supplied configuration.
- The warranty expires if the product is incorrectly assembled, not used as intended and/or handled improperly.
- Make sure at all times that the diaphragm-type accumulator is positioned safely and secured against unintended rolling or falling.
- Do not expose the product to any mechanical loads under any circumstances:
 - Do not place any objects on top of it.
 - Never use the product as handle or step.
 - Do not apply further loads.

During transport

- Diaphragm-type accumulators may fall down during transport. Due to their weight, this results in the risk of considerable injuries.
- When diaphragm-type accumulators with a capacity of 2.0 l and more are lifted, there is the risk of health hazards due to their weight. Use adequate lifting tools for the transport.

General safety instructions

- During assembly**
- Always depressurize and de-energize the relevant plant part before assembling the diaphragm-type accumulator. Secure the system against restarting. In this connection, always observe the operating instructions and any other technical documentation relating to the system.
 - Do not carry out any welding, soldering or mechanical works on the diaphragm-type accumulator.
 - Before assembling the diaphragm-type accumulator, check it for transport damages.
 - Make sure that the seal and the sealing surface of the fluid connection are intact.
 - During assembly, ensure utmost cleanliness in order to avoid that dirt may penetrate the hydraulic lines and lead to wear, malfunctions or a complete failure.
 - Allow the product to acclimatize for some hours prior to assembly, as otherwise condensed water may be generated.
 - Secure the diaphragm-type accumulator against unintended rolling or falling during the assembly in order to avoid any personal injuries or material damage to the diaphragm-type accumulator.
 - Before mounting the hydraulic accumulator to the fluid connection, make sure that the connection variant on the side of the hydraulic accumulator corresponds to the connection on the system side.
 - Fasten the diaphragm-type accumulator safely by means of mounting clamps according to 50150 in order to relieve the pipe union and in order to avoid loosening or damage due to vibrations.
 - Immediately wipe off any hydraulic fluid leaking during assembly in order to avoid a slip hazard.
- During commissioning**
- Make sure that the system is depressurized before charging the diaphragm-type accumulator with nitrogen.
 - Only charge the diaphragm-type accumulator with nitrogen 99.99 vol. %. If the diaphragm-type accumulator is charged with other gases, particularly with air or oxygen, the diaphragm-type accumulator might explode.
 - For charging the diaphragm-type accumulator with nitrogen, always use the charging and testing device according to data sheet 50150.
 - After charging the accumulator, mark the operating pressure permanently and clearly visible on the diaphragm-type accumulator. In this connection, do not use stamps, punch presses or other mechanical markings.
 - Commission the product only if it is installed completely.
 - Observe the operating instructions of the machine/system.
- During operation**
- Only authorized staff is allowed to operate the adjustment elements at components or parts, under the provision that the hydraulic system is used as intended.
 - Only persons who have been authorized by the operator may be granted access to the direct operating range. This also applies when the system is at standstill.
 - Make sure that the embossing on the housing is always present and legible. Otherwise, further operation is not allowed.
 - In case of emergency, an error, or in case of other irregularities, switch off the system and secure it against restarting.
 - The diaphragm-type accumulator must not be operated with corrosive hydraulic fluid. Otherwise, there is a risk of reduction of wall thickness and thus a danger of bursting for the pressure vessel.
 - Make sure that no forces are applied to the diaphragm-type accumulator. The effects of external forces may lead to damages on the fluid connection and the uncontrolled release of pressurized hydraulic fluids.

General safety instructions

- Make sure that the ambient temperature of the diaphragm-type accumulator does not exceed or fall below the ambient temperature indicated in the technical data and on the embossing.
- During cleaning**
- Close the fluid connection and cover the gas port by means of the cover caps provided.
 - Never use solvents or aggressive detergents.
 - Only clean the diaphragm-type accumulator using a dry, lint-free cloth.
 - Do not use a pressure washer for cleaning.
- During maintenance and repairs**
- The operator is obliged to subject the diaphragm-type accumulator to regular controls according to locally applicable provisions. Check which provisions or ordinances are to be complied with in your country and/or for your application.
 - Perform the prescribed maintenance works at the intervals that are specified in the operating instructions of the overall system. If no other instructions are available, we recommend maintenance intervals according to chapter "9.1 Maintenance".
 - Make sure that no lines connections, ports and components are disconnected as long as the system is under pressure and voltage or the diaphragm-type accumulator is still under hydraulic pressure. Secure the system against restarting.
 - Secure the diaphragm-type accumulator against unintended falling or rolling.
 - For the discharging of the diaphragm-type accumulator, always use the charging and testing device according to data sheet 50150. The uncontrolled release of charging gas results in a risk of suffocation and a risk of injury due to entrained components. During discharging, provide for sufficient ventilation of the working space.
 - For the charging of the diaphragm-type accumulator, always use the charging and testing device according to data sheet 50150.
 - Only charge the diaphragm-type accumulator with nitrogen 99.99 vol. %. If the diaphragm-type accumulator is charged with a gas not complying with the intended use, the diaphragm-type accumulator might explode.
 - After completion of the repair works, make sure that the fluid connection is correctly connected again and the gas port is covered.
- During decommissioning**
- Make sure that no lines connections, ports and components are disconnected as long as the system is under pressure and voltage or the diaphragm-type accumulator is still under hydraulic pressure. Secure the system against restarting.
 - For the discharging of the diaphragm-type accumulator, always use the charging and testing device according to data sheet 50150. The uncontrolled release of charging gas results in a risk of suffocation and a risk of injury due to entrained components. During discharging, provide for sufficient ventilation of the working space.
 - When diaphragm-type accumulators with a capacity of 2.0 l and more are disassembled, there is the risk of health hazards due to their weight. Use adequate lifting tools during disassembly.
 - Put down the diaphragm-type accumulator safely and secure it against unintended rolling or falling in order to avoid any injuries or damages.
 - Avoid a reuse of the diaphragm-type accumulator as pressure vessel by sawing it open or drilling through it at the welded seam.
- Disposal**
- Dispose of the hydraulic fluid according to the provisions valid in your country as well as according to the safety data sheets of the hydraulic fluids.

2.5 Obligations of the operator

The diaphragm-type accumulator is supplied ex works with a black UV coating, UE16-M9502-2 painted. The operator is responsible for assuring a sufficient corrosion protection corresponding to the ambient conditions and requirements.

The operator of the Bosch Rexroth products is bound to provide for personnel training on a regular basis regarding the following subjects:

- Observation and use of the operating instructions and the legal regulations
- Intended use of the product
- Observation of the instructions on occupational safety and operating instructions

3 Scope of delivery

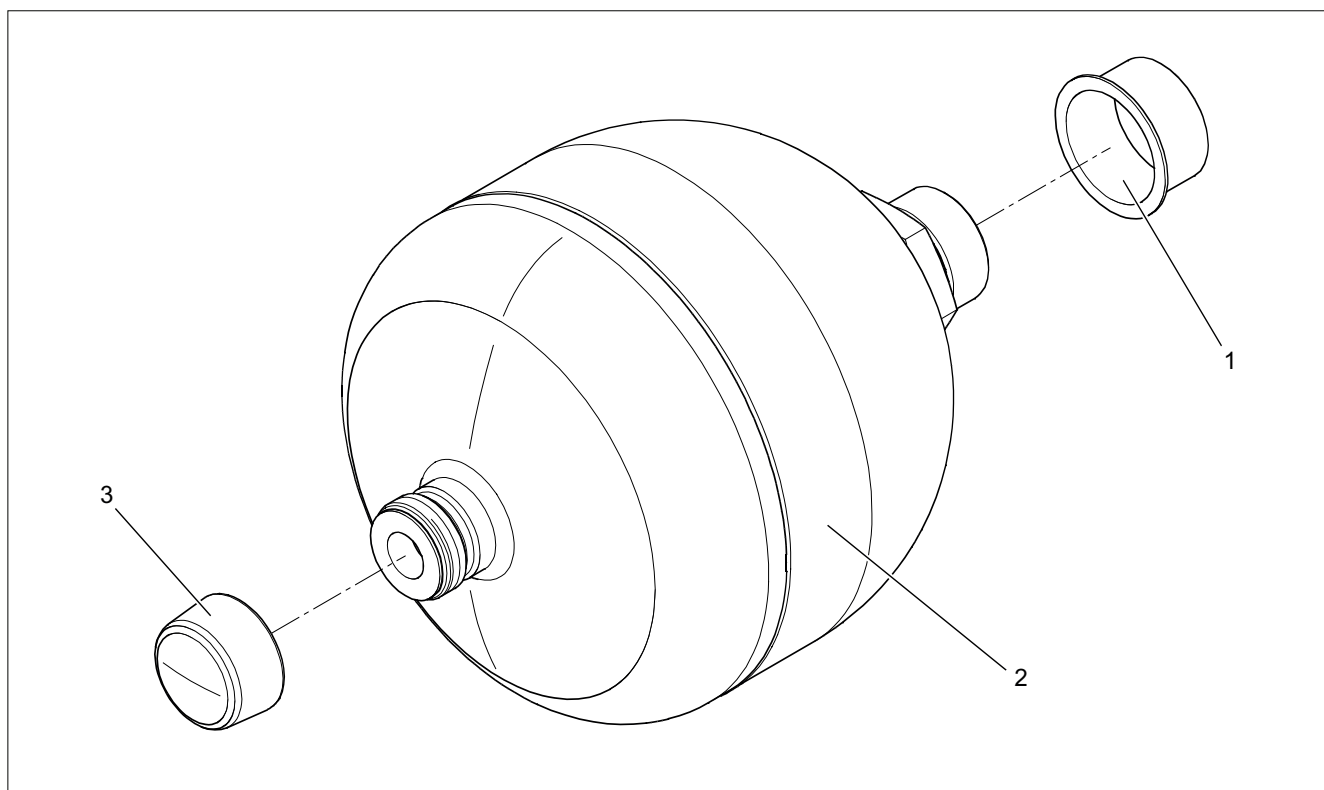


Fig. 1: Scope of delivery HAD..1X/-2X diaphragm-type accumulator

The scope of delivery includes:

- HAD..1X/-2X diaphragm-type accumulator (2), depending on the quantity ordered
- 1 operating manual in German and English language
- 1 declaration of conformity (only for diaphragm-type accumulators having a capacity of more than 1 l)

The diaphragm-type accumulator is supplied with the following parts being mounted (optional, according to supplied version):

- Cover cap on the gas port (3)
- Protective cap on the fluid connection (1)

4 Product description

4.1 Performance and device description

Diaphragm-type accumulators are intended for the use in hydraulic systems. They are used for energy storage, shock and vibration absorption, and leakage oil compensation or volume compensation. The high compressibility of gases is used for the diaphragm-type accumulator.

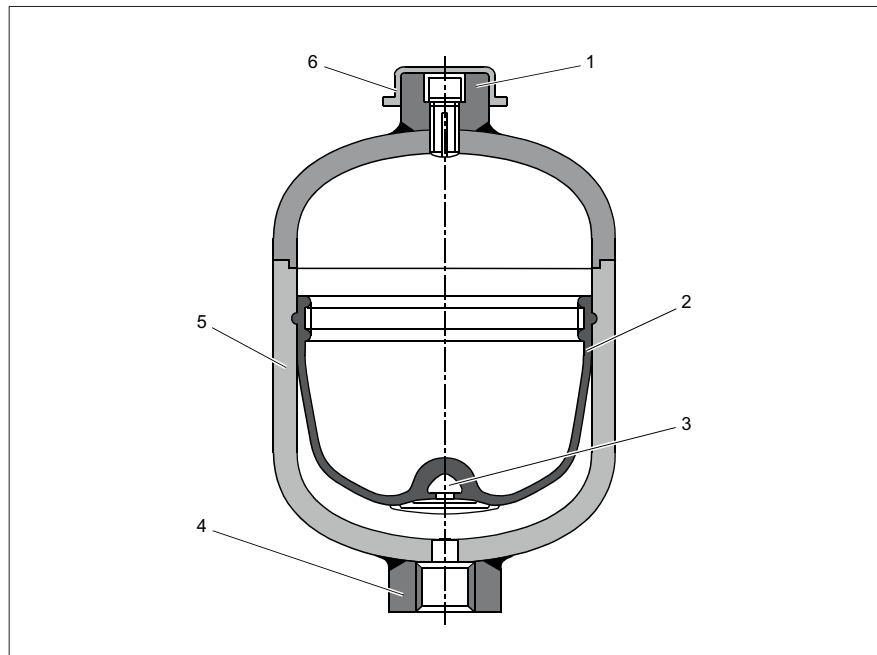


Fig. 2: Structure of the diaphragm-type accumulator

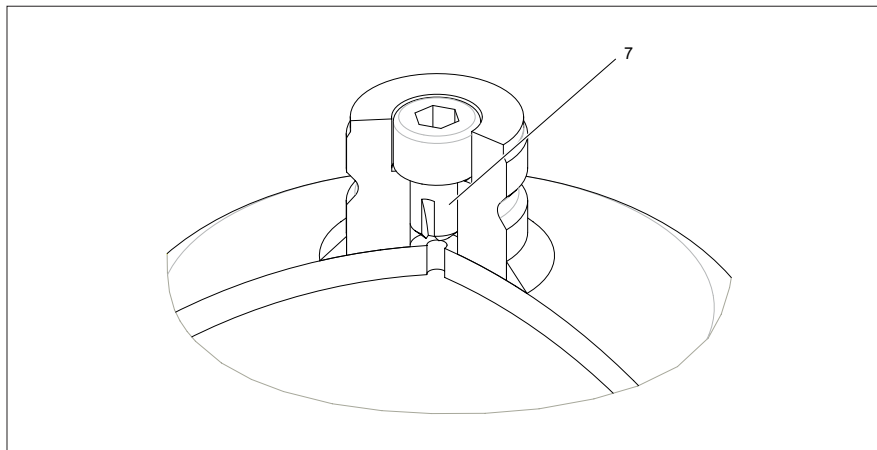


Fig. 3: Structure of the gas port

Diaphragm-type accumulators consist of a pressure-resistant vessel (5) made of high-tensile steel which most commonly has a spherical or cylindrical form.

The hydraulic accumulator is subdivided into a gas and a fluid side by a diaphragm (2) made of an elastic, flexible material (elastomer).

When the operating pressure is increased, hydraulic fluid is flowing into the diaphragm-type accumulator and compresses the gas until the gas pressure is identical to the fluid pressure. When the operating pressure is reduced, the gas expands again and feeds the hydraulic system with fluid.

The closing poppet (3) located at the bottom of the diaphragm completely covers the fluid connection (4) when the diaphragm-type accumulator is discharged, thus preventing the diaphragm from exiting into the fluid channel and being damaged.

Via the gas port (1), the gas side of the diaphragm-type accumulator is charged with nitrogen to the provided pre-filling pressure p_0 by loosening the gas filling screw (7). In order to protect the gas port, a cover cap (6) is provided on the gas port.

More detailed information on operating conditions, connection dimensions and limits of performance are provided in the data sheet 50150.

The weight depends on the vessel size and the maximum operating pressure, see "Table 4: Weight indications".

Table 4: Weight indications

Type/V in liters	p_{max} in bar	Weight in kg
HAD 0.075	250	0.65
HAD 0.16	250	1.0
HAD 0.35	210	1.3
HAD 0.5	160	1.6
	250	2.0
HAD 0.7	180	2.6
	250	3.2
HAD 1.0	200	3.5
HAD 1.4	140	4.9
	250	6.2
	350	7.7
HAD 2.0	100	4.0
	250	9.5
	350	11.9
HAD 2.8	70	5.5
	250	10.0
	350	15.6
HAD 3.5	250	14.0
	350	18.0

4.2 Identification of the diaphragm-type accumulator

The diaphragm-type can be identified by means of its embossing on the housing. The operation of the diaphragm-type accumulator is only admissible, when the embossing is present and fully legible.

The following example shows the embossing of a diaphragm-type accumulator:

Product description

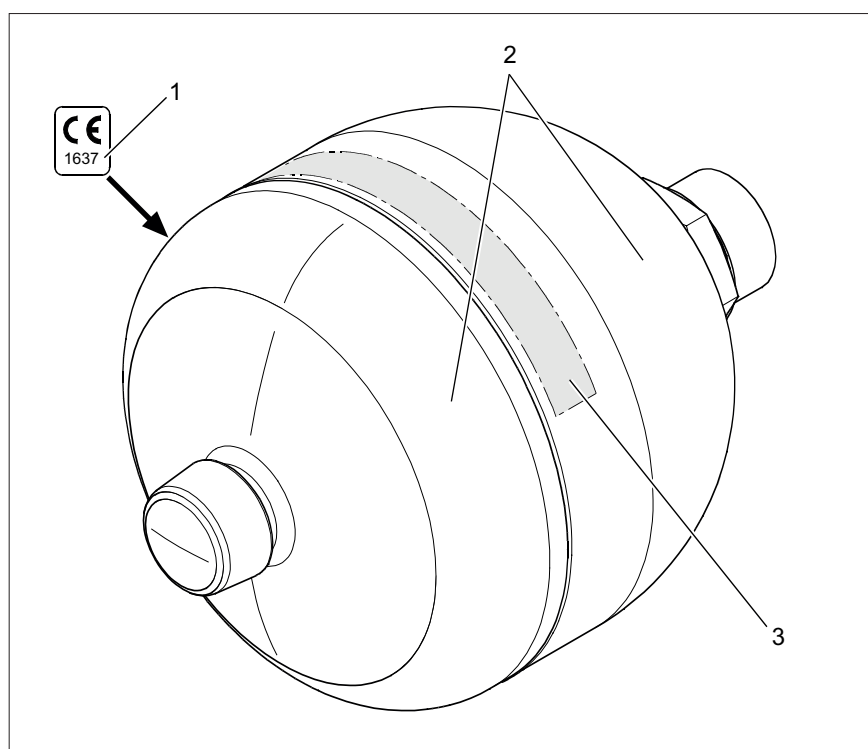


Fig. 4: Position of diaphragm-type accumulator marking

- | | |
|---|---|
| <p>1 CE mark (only for diaphragm-type accumulators having a capacity of more than 1 l)</p> <p>2 Marking of material batch</p> | <p>3 Embossing on the housing of the diaphragm-type accumulator</p> |
|---|---|

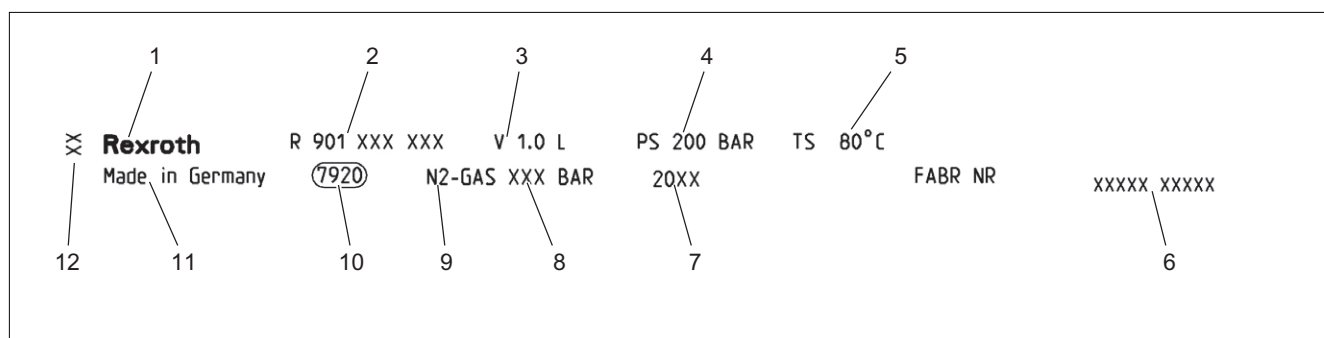


Fig. 5: Example of embossing on the housing of the diaphragm-type accumulator

- | | |
|--|---|
| <p>1 Brand</p> <p>2 Material number</p> <p>3 Effective gas volume</p> <p>4 Maximum operating pressure</p> <p>5 Maximum operating temperature</p> <p>6 Serial number / manufacturing week</p> | <p>7 Manufacturing year</p> <p>8 Pre-filling pressure ex works</p> <p>9 Charging gas</p> <p>10 Area / works number</p> <p>11 Designation of origin</p> <p>12 Factory inspectorate</p> |
|--|---|

5 Transport and storage

5.1 Transporting the diaphragm-type accumulator

CAUTION!



Risk of health hazards!

When diaphragm-type accumulators with a capacity of 2.0 l and more are lifted, there is the risk of health hazards due to their heavy weight.

- ▶ Use adequate lifting tools for the transport.

WARNING!



Crush injuries and fractures!

Due to the round external contours, diaphragm-type accumulators may roll away or fall down, as the case may be. This may cause serious personal injuries and damage to property.

- ▶ Make sure that the diaphragm-type accumulator is positioned safely and secured against unintended rolling or falling.

CAUTION!



Risk of damage!

Impact or similar forces may damage the diaphragm-type accumulator.

- ▶ Do not apply mechanical forces to the product.
- ▶ Do not apply any further external loads.
- ▶ Never use the product as handle or step.
- ▶ Do not place any objects on top of it.

CAUTION!



Risk of damage!

Transporting the diaphragm-type accumulator on the gas port may cause damage to the accumulator.

- ▶ Never lift the diaphragm-type accumulator on the gas port.
- ▶ Always use adequate lifting tools for the transport.

Close openings

- ▶ Before the transport, close all openings by means of appropriate protective caps/devices in order to prevent dirt or water from penetrating into the diaphragm-type accumulator.

Transport

Diaphragm-type accumulators may be transported by means of a forklift or adequate lifting tools.

- ▶ Use adequate lifting tools for diaphragm-type accumulators with a capacity of 2.0 l and more.

Weight

The weight indicated only refers to the diaphragm-type accumulator, the weight of any add-on units is not taken into account. Information on the weight of the diaphragm-type accumulator are indicated in Table 4 on page 15.

Further transport

If the diaphragm-type accumulator is transported with a preload pressure of 2 bar and more, it must be declared as hazardous material. When transporting the diaphragm-type accumulator by train, ship or air freight, the relevant directives are to be observed. When transporting the accumulator by air freight, a dispatch at zero pressure is recommended.

5.2 Storing the diaphragm-type accumulator

The diaphragm-type accumulator is supplied with a coating; the operator must ensure a sufficient corrosion protection during storage.

Requirements

- The storage facilities must be free from etching substances and gases.
- The storage facilities must be dry.
- The ideal storage temperature ranges from +5 °C to +20 °C.
- The temperature should ideally be constant.

Maximum storage time

The maximum storage time of the diaphragm-type accumulator amounts to 5 years. Procedure after expiration of maximum storage time:

- Carry out a visual examination of the diaphragm-type accumulator and check for any damage and corrosion.
- After the expiration of the maximum storage time, have the complete diaphragm-type accumulator checked by a competent Rexroth service.



Please note that the warranty period is not prolonged by the storage time!



Please note that, according to applicable provisions and ordinances, test intervals mostly refer to the date of manufacture and are also not prolonged by the storage time.

6 Assembly

The assembly must only be carried out after the hydraulic scheme of the system or an assembly drawing is available.

6.1 Unpacking

CAUTION!



Danger from falling parts!

If its packaging is opened improperly, the diaphragm-type accumulator may roll out and cause injuries or damage.

- Put the packaging on a level bearing surface.

1. Before opening the packaging and/or loosening the tension belts, make sure that the accumulator/s cannot roll away.
2. Open the packaging and/or loosen the tension belts.
3. Use adequate lifting tools when lifting a diaphragm-type accumulator with a capacity of 2.0 l or more out of the packaging.
4. Check the diaphragm-type accumulator for obvious transport damages and/or other damages.
5. Dispose of the packaging in accordance with the provisions applicable in your country.

6.2 Installation conditions

Fastening	Hydro-pneumatic accumulators with a capacity of 2.0 l or more must be sufficiently secured and fixed due to their heavy weight and the acceleration forces generated by the hydraulic fluid in the hydraulic accumulator. The accumulator is to be fixed in such a way that a safe support during operation-related vibrations or in case of a potential pipeline break is guaranteed. Bosch Rexroth offers corresponding retaining clamps, see data sheet 50150.
Protection	According to the Pressure Equipment Directive 97/23/EC, the diaphragm-type accumulator must be secured against excessive pressures.
Cleanness	It is imperative to provide for absolute cleanliness. The diaphragm-type accumulator and all other parts used must be installed free from dirt. Contamination of the hydraulic fluid may considerably impair the service life of the diaphragm-type accumulator.
Cleaning	Clean the accumulator with dry, lint-free cleaning cloths.
Temperature	The temperature of the diaphragm-type accumulator must correspond to the ambient temperature of the installation site. Allow the diaphragm-type accumulator to acclimatize sufficiently in order to adapt to the temperature conditions.
Installation position	The diaphragm-type accumulator may be installed in any installation position. Above the gas port, a clearance area of 200 mm must be observed in order to be able to connect the charging and testing devices.
Ports	Before mounting the hydraulic accumulator to the fluid connection, make sure that the connection variant on the side of the hydraulic accumulator corresponds to the connection on the system side.

6.3 Required tools

For the assembly of the diaphragm-type accumulator, you need:

- Charging and testing device according to data sheet 50150
- Open-end wrench - for information on wrench sizes see data sheet 50150, according to fluid connection variant
- Adequate lifting tools for volumes of 2.0 l and more

6.4 Assembling the diaphragm-type accumulator

CAUTION!



Danger of damage to property and personal injuries!

The assembly of the diaphragm-type accumulator requires basic mechanical and hydraulic knowledge.

- The diaphragm-type accumulator may only be assembled by qualified personnel (see chapter "2.3 Qualification of personnel")

CAUTION!



Risk of damage!

Impact or similar forces may damage the diaphragm-type accumulator.

- Do not apply mechanical forces to the product.
- Never use the product as lever, handle or step.
- Do not place any objects on top of it.

Assembly

CAUTION!**Danger of bursting!**

Bursting of the diaphragm-type accumulator due to welding, soldering or other mechanical works may lead to serious injuries.

- ▶ Do not carry out any mechanical works such as welding or soldering on the diaphragm-type accumulator.

CAUTION!**Risk of damage!**

Transporting the diaphragm-type accumulator on the gas port may cause damage to the accumulator.

- ▶ Never lift the diaphragm-type accumulator on the gas port.
- ▶ Always use adequate lifting tools for the transport.

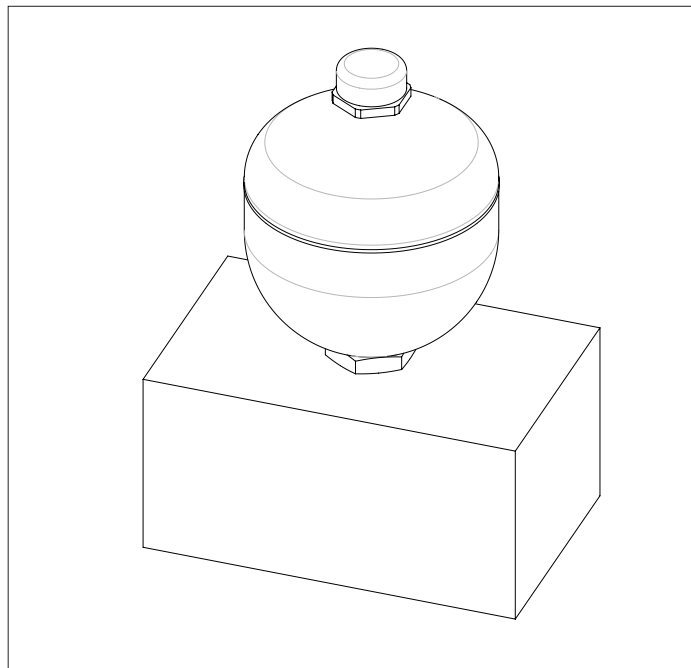
6.4.1 Installing the diaphragm-type accumulator on a block

Fig. 6: Diaphragm-type accumulator installed on a block

For installing the diaphragm-type accumulator on a manifold block, the following work steps are required:

WARNING!**Risk of poisoning and injuries due to leaking hydraulic fluid!**

The contact with hydraulic fluids leads to health hazards (e.g. eye injuries, skin lesions, poisoning during installation).

Slipping may cause serious injuries. When covers are removed, unpressurized residual hydraulic fluid may leak.

- ▶ Wear protective gloves, safety goggles and suitable working clothes.
- ▶ Immediately wipe off any leaking oil.
- ▶ Check the lines for wear and damages before any commissioning.
- ▶ If hydraulic fluid comes into contact with the eyes or penetrates the skin nevertheless, please consult a doctor immediately.

1. Remove the cover cap on the fluid connection of the diaphragm-type accumulator.
2. Make sure that the fluid connection is clean and free from foreign matter.
3. Ensure that the connection thread is intact.
4. Check the fluid connection for available seals.

CAUTION!**Crush injuries and fractures!**

Falling or rolling diaphragm-type accumulators may cause serious injuries.

- ▶ Use adequate lifting tools when lifting a diaphragm-type accumulator with a capacity of 2.0 l and more.
- ▶ Secure the diaphragm-type accumulator against unintended rolling.

-
5. Fix diaphragm-type accumulators with a capacity of 2.0 l and more by means of adequate lifting tools so that a safe support is guaranteed.
 6. Place the diaphragm-type accumulator on the oil port.
 7. Manually screw the accumulator into the thread of the oil port.
 8. Make sure that the thread interlocks after one rotation at the latest.
 9. If need be, lower the accumulator and manually screw it into the thread until the stop of the thread is reached. The end face of the port must fit completely to the counterpart.
 10. Continue lowering the lifting device and make sure that the accumulator is firmly seated on the oil port.
 11. Tighten the accumulator by means of a suitable tool until it fits tightly.
 12. Remove the lifting device if required.

The diaphragm-type accumulator is now mounted.

6.4.2 Mounting the diaphragm-type accumulator by means of retaining clamp

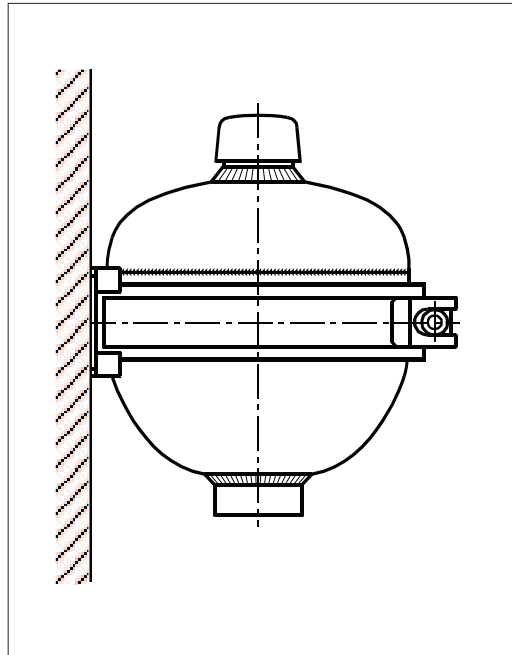


Fig. 7: Mounting by means of retaining clamp

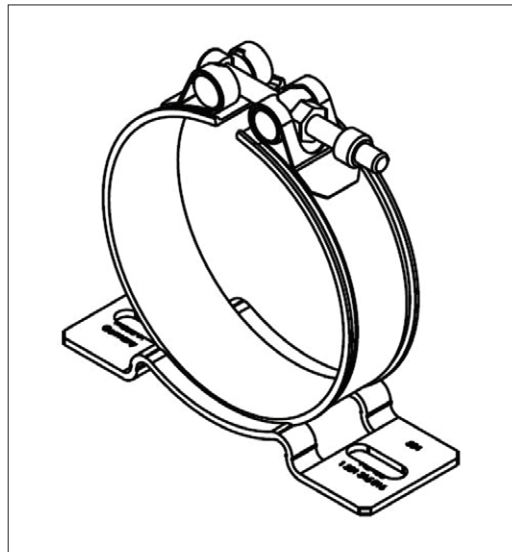


Fig. 8: Retaining clamp

For mounting the diaphragm-type accumulator by means of a retaining clamp, the following work steps are required:

1. Mount the holding device of the diaphragm-type accumulator to the provided positions on the wall.

CAUTION!**Crush injuries and fractures!**

Falling or rolling diaphragm-type accumulators may cause serious injuries.

- ▶ Use adequate lifting tools when lifting a diaphragm-type accumulator with a capacity of 2.0 l and more.
- ▶ Secure the diaphragm-type accumulator against unintended rolling.

-
2. Lift diaphragm-type accumulators with a capacity of 2.0 l and more using adequate lifting tools and position the accumulator within the retaining clamp.
 3. Fasten the retaining clamps at the provided points.
 4. Remove the lifting device.

WARNING!**Risk of poisoning and injuries due to leaking hydraulic fluid!**

The contact with hydraulic fluids leads to health hazards (e.g. eye injuries, skin lesions, poisoning during installation).

Slipping may cause serious injuries. When covers are removed, unpressurized residual hydraulic fluid may leak.

- ▶ Wear protective gloves, safety goggles and suitable working clothes.
- ▶ Immediately wipe off any leaking oil.
- ▶ Check the lines for wear and damages before any commissioning.
- ▶ If hydraulic fluid comes into contact with the eyes or penetrates the skin nevertheless, please consult a doctor immediately.

-
5. Remove the cover cap on the fluid connection of the diaphragm-type accumulator.
 6. Connect the hydraulic lines to the fluid connection of the diaphragm-type accumulator by exerting counterpressure with an open-end wrench on the oil valve.
 7. Tighten the hydraulic lines by exerting counterpressure with a suitable tool on the oil valve until they fit tightly.
 8. Ensure that the diaphragm-type accumulator is mounted without tension stress and check if any tension stress may occur due to vibrations and/or thermal expansion of the components and/or the piping.
 9. Remove the lifting device if required.

The diaphragm-type accumulator is now mounted.

7 Commissioning

DANGER!



Danger of damage to property and personal injuries!

The commissioning of the diaphragm-type accumulator requires basic mechanical and hydraulic knowledge.

- ▶ The diaphragm-type accumulator may only be commissioned by qualified personnel (see chapter "2.3 Qualification of personnel").

WARNING!



Danger of damage to property and personal injuries!

If the diaphragm-type accumulator is not mounted correctly, persons could be injured or the product or system could be damaged when commissioning the diaphragm-type accumulator.

- ▶ Make sure that the diaphragm-type accumulator has been mounted correctly by qualified personnel before commissioning the diaphragm-type accumulator.

CAUTION!



Damage to property!

Polluted hydraulic fluid could result in wear and malfunctions.

- ▶ During commissioning, absolute cleanliness must be ensured.
- ▶ Make sure that the hydraulic system is only filled with a filtered hydraulic fluid.
- ▶ Ensure that no pollutants are able to penetrate when sealing the ports.

Concerning the commissioning of the diaphragm-type accumulator, please comply with the operating instructions of the hydraulic system.

For the commissioning of the diaphragm-type accumulator, you need the Bosch Rexroth charging and testing device according to data sheet 50150.

7.1 Preparing for commissioning

The diaphragm-type accumulator is supplied ex works with a black UV coating, UE16-M9502-2 painted. The operator is responsible for assuring a sufficient corrosion protection corresponding to the ambient conditions and requirements.

The examination prior to the commissioning as well as the recurring tests are to be carried out according to national rules.

- ▶ Add the coating at the gas port, if required.
- ▶ Make sure that the diaphragm-type accumulator is intact.
- ▶ By means of the information provided on the embossing on the housing, make sure that the diaphragm-type accumulator corresponds to the specifications provided on the hydraulic scheme or in the system parts list.
- ▶ Check that the fluid used in the hydraulic system corresponds to the specifications provided in the technical data sheet of the diaphragm-type accumulator.
- ▶ Make sure that the maximum operating pressure of the diaphragm-type accumulator is equal to or greater than the maximum operating pressure of the hydraulic circuit.
- ▶ Check whether the operating temperatures are within the limits mentioned on the embossing.
- ▶ Lay out ready the Bosch Rexroth charging and testing device according to data sheet 50150 and its operating instructions.

7.2 First commissioning

In order to commission the HAD..1X/-2X diaphragm-type accumulator, proceed as follows:

Setting the pre-filling pressure

Whenever the diaphragm-type accumulator is commissioned, it must be brought to the prescribed pre-filling pressure according to the scheme.

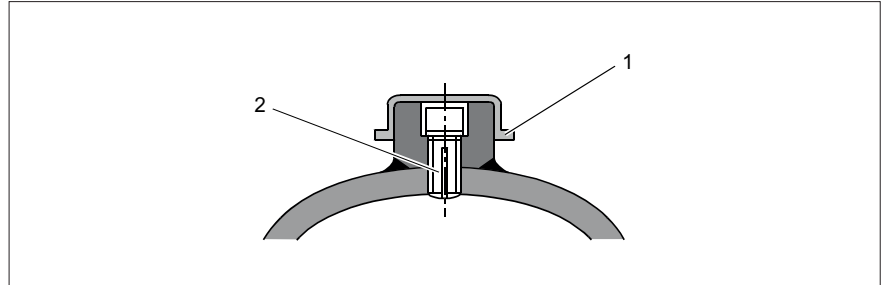


Fig. 9: Inspection of the pre-filling pressure

1 Protective cap

2 Gas filling screw

To adjust the pre-filling pressure, proceed as follows:

1. Make sure that the system is depressurized.

DANGER!



Risk of injury and suffocation!

Risk of injury due to uncontrolled release of a greater volume of gas and risk of injury due to entrained components.

- Make sure that the workplace is sufficiently ventilated.
- For checking the pre-filling pressure, always use a suitable charging and testing device.

2. Remove the protective cap from the gas side of the diaphragm-type accumulator.
3. Screw the prefill and test valve for hydraulic accumulators onto the gas port of the diaphragm-type accumulator. When doing this, always observe the operating instructions of the charging and testing device.

DANGER!



Risk of explosion!

Risk of serious injuries due to explosion of the hydraulic accumulator during the charging of the diaphragm-type accumulator with improper gas.

- Only charge the diaphragm-type accumulator with nitrogen 99.99 vol. %.

4. Connect the nitrogen bottle to the prefill valve via the hose of the charging and testing device.
5. The prescribed pre-filling pressure is indicated in the hydraulic scheme.

Commissioning

CAUTION!**Damage to property!**

If the pre-filling pressure is too high, this may cause damage to the diaphragm of the accumulator.

- ▶ Observe the maximum pre-filling pressure of 130 bar.

6. Open the stop cock of the gas bottle.
7. Loosen the gas filling screw via the prefill and test valve until gas can flow into the accumulator.
8. Charge the diaphragm-type accumulator until the prescribed pre-filling pressure is indicated on the pressure gauge of the prefill valve. Always keep an eye on the pressure gauge during the charging process.
9. Close the stop cock of the gas bottle.
10. The pre-filling pressure depends on the temperature. During the charging, the diaphragm-type accumulator heats up. Wait until the accumulator cools down.
11. Check the pre-filling pressure and correct it if necessary.
12. Tighten the gas filling screw at the diaphragm-type accumulator again via the prefill and test valve using a tightening torque of 25 Nm.
13. Remove the prefill and testing valve.
14. Screw the protective cap back on the accumulator.
15. Mark the prescribed pre-filling pressure visibly on the diaphragm-type accumulator.

The pre-filling pressure is now checked and set.

Commissioning the diaphragm-type accumulator

- ▶ Commission the diaphragm-type accumulator with the system according to the operating instructions of the system.

7.3 Re-commissioning after extended standstill

- ▶ Check whether the pre-filling pressure corresponds to the value p_0 indicated on the embossing on the housing. For this, proceed as described in chapter "9.1 Maintenance".
- ▶ Check the connection thread for leak-tightness.
- ▶ Make sure that the accumulator does not show any signs of corrosion and that the coating is intact.
- ▶ Commission the diaphragm-type accumulator with the system according to the operating instructions of the system.

8 Operation

The HAD..1X/-2X diaphragm-type accumulator must only be operated within the defined limits of performance, see the technical data sheet of the diaphragm-type accumulator 50150.

The diaphragm-type accumulator may only be operated if it is in an unobjectionable condition.

In order to guarantee that the diaphragm-type accumulator may be operated reliably for a long period of time, Bosch Rexroth recommends to regularly check and maintain the hydraulic system and the diaphragm-type accumulator. Please observe the information provided in chapter "9.1 Maintenance" as well as the maintenance regulations of the system manufacturer.

Changes in operating speeds, temperatures, increasing noises or power consumption are an indication of wear or damage at the accumulator or the system. In order to guarantee a high operational safety and availability of the system, it is recommended to continuously monitor these values.

9 Maintenance and repair



No warranty is accepted for independently performed repairs!

9.1 Maintenance

WARNING!



Risk of poisoning and injuries due to leaking hydraulic fluid!

The contact with hydraulic fluids leads to health hazards (e.g. eye injuries, skin lesions, poisoning during installation).

Slipping may cause serious injuries. When covers are removed, unpressurized residual hydraulic fluid may leak.

- ▶ Wear protective gloves, safety goggles and suitable working clothes.
- ▶ Immediately wipe off any leaking oil.
- ▶ Check the lines for wear and damages before any commissioning.
- ▶ If hydraulic fluid comes into contact with the eyes or penetrates the skin nevertheless, please consult a doctor immediately.

DANGER!



Risk of injury and suffocation!

Risk of injury due to uncontrolled release of a greater volume of gas and risk of injury due to entrained components.

- ▶ For checking the pre-filling pressure, always use a suitable charging and testing device.

DANGER!



Risk of explosion!

Risk of serious injuries due to explosion of the hydraulic accumulator during the charging of the diaphragm-type accumulator with improper gas.

- ▶ Only charge the diaphragm-type accumulator with nitrogen 99.99 vol. %.

Maintenance and repair

DANGER!**Crush injuries and fractures!**

Falling or rolling diaphragm-type accumulators may cause serious injuries.

- ▶ Use adequate lifting tools when lifting a diaphragm-type accumulator with a capacity of 2.0 l and more.
- ▶ Secure the diaphragm-type accumulator against unintended rolling.

WARNING!**Risk of injury!**

Parts flying around due to the residual pressure that is still present after the discharging may lead to serious injuries.

- ▶ Check the residual pressure in the diaphragm-type accumulator after heating it up to the ambient temperature, before opening the gas filling screw.

Checking the pre-filling pressure

Maintenance activities for the diaphragm-type accumulator:

To check the pre-filling pressure of the diaphragm-type accumulator, proceed as follows:

1. Make sure that the system is depressurized.
2. The prescribed pre-filling pressure is indicated in the hydraulic scheme.

CAUTION!**Risk of injury and suffocation!**

Risk of injury due to uncontrolled release of a greater volume of gas and risk of injury due to entrained components.

- ▶ Make sure that the workplace is sufficiently ventilated.
- ▶ For checking the pre-filling pressure, always use a suitable charging and testing device.

3. Remove the protective cap from the gas side of the diaphragm-type accumulator.
4. Screw the prefill and test valve for hydraulic accumulators onto the gas port of the diaphragm-type accumulator. When doing this, always observe the operating instructions of the charging and testing device.
5. The prescribed pre-filling pressure is indicated in the hydraulic scheme.
6. Continue loosening the gas filling screw at the diaphragm-type accumulator until the pre-filling pressure is indicated on the pressure gauge.
7. Check whether the pre-filling pressure indicated on the pressure gauge of the charging and testing device corresponds to the value indicated in the hydraulic scheme. If the two values differ, set the pre-filling pressure as described in chapter "7.2 First commissioning".
8. Tighten the gas filling screw on the diaphragm-type accumulator again.
9. Remove the prefill and test valve for hydraulic accumulators from the gas port of the diaphragm-type accumulator.

The pre-filling pressure is now checked and set.

Inspections After having been charged with gas, Bosch Rexroth hydraulic accumulators are almost maintenance-free. In order to guarantee a failure-free operation and a long service life, however, the following tasks are to be carried out:

- Inspection of the pre-filling pressure
- Inspection of the safety equipment, fittings
- Inspection of the line connections
- Inspection of accumulator fixation

If the accumulator is to be disassembled for maintenance tasks, the accumulator must be depressurized first.

Test intervals

Table 5: Test intervals for the diaphragm-type accumulator

Test	Interval	Maintenance activities
Inspection of the pre-filling pressure with external visual examination	Test 1: Within one week as of commissioning	Inspection of the pre-filling pressure; inspection of tightness, connection thread; visual examination of corrosion protection.
	Test 2: Within 3 months as of commissioning, if no gas losses during test 1	
	Test 3: Annual test, if no gas losses during test 2	
Internal visual examination	E. g. once every ten years	Inspection of the pressure vessel according to national provisions. If this test is to be carried out due to national provisions, this is not possible due to the design. The accumulator is to be replaced.

Maintenance schedule of the system

For a safe operation and a long service life of the diaphragm-type accumulator, a maintenance schedule is to be drawn up for the power unit, the machine, or the system. The maintenance schedule must guarantee that the operating conditions of the diaphragm-type accumulator stay in the prescribed limits during the entire storage time.

In particular, compliance with the following operating parameters has to be ensured:

- Operating pressure
- Operating temperature range
- Surface temperature
- External pressure

Changes made to these parameters increase the wear of the diaphragm-type accumulator. The cause has to be determined and remedied immediately.

In order to guarantee a high operational safety of the diaphragm-type accumulator in the machine/system, we recommend continuous automatic controls of the parameters mentioned above.

Further information on the maintenance is provided in the operating instructions of the system.

9.2 Repair

The diaphragm-type accumulator is not repairable. In the case of damages, an adequate replacement is to be provided.

9.3 Spare parts

There are no spare parts available for the diaphragm-type accumulator. In case of questions concerning the replacement procurement, please contact the competent Rexroth service or the central service department:

Bosch Rexroth AG
Service Industriehydraulik [Industrial hydraulics]
Competence Center Lohr
Bgm.-Dr.-Nebel-Str. 8
97816 Lohr am Main
Phone: +49 (0) 93 52 - 18 0
spareparts.bri@boschrexroth.de

The addresses of our sales and service network are provided under:
www.boschrexroth.com/adressen

10 Disassembly and replacement

- ▶ Discharge the accumulator via the discharge device of the system.
Please note that the pre-filling pressure still exists on the gas side after hydraulic relief.
- ▶ Please decide whether a remaining gas pressure in the diaphragm-type accumulator is required or reasonable.
- ▶ For safety reasons, we recommend discharging the gas pressure and clearly marking this on the accumulator, if the system is shut-down for a longer time or forever.
- ▶ In order to keep the accumulator permanently depressurized, the gas filling screw must be removed.

10.1 Required tools

For the disassembly of the diaphragm-type accumulator, you need:

- Charging and testing device according to data sheet 50150
- Open-end wrench - for information on wrench sizes see data sheet 50150, according to fluid connection variant
- Adequate lifting tools for volumes of 2.0 l and more

10.2 Preparing disassembly

Decommission the overall system as described in the operating instructions of the system.

Afterwards, prepare the disassembly of the diaphragm-type accumulator as follows:

- ▶ Depressurize the hydraulic system.
- ▶ Make sure that the relevant plant parts are de-pressurized and de-energized.

10.3 Disassembling the diaphragm-type accumulator

Proceed as follows to disassemble the diaphragm-type accumulator:

WARNING!



Risk of poisoning and injuries due to leaking hydraulic fluid!

The contact with hydraulic fluids leads to health hazards (e.g. eye injuries, skin lesions, poisoning during installation).

Slipping may cause serious injuries. When covers are removed, unpressurized residual hydraulic fluid may leak.

- ▶ Wear protective gloves, safety goggles and suitable working clothes.
- ▶ Immediately wipe off any leaking oil.
- ▶ Check the lines for wear and damages before any commissioning.
- ▶ If hydraulic fluid comes into contact with the eyes or penetrates the skin nevertheless, please consult a doctor immediately.

DANGER!



Crush injuries and fractures!

Falling or rolling diaphragm-type accumulators may cause serious injuries.

- ▶ Use adequate lifting tools when lifting a diaphragm-type accumulator with a capacity of 2.0 l and more.
- ▶ Secure the diaphragm-type accumulator against unintended rolling.

WARNING!



Risk of injury!

Parts flying around due to the residual pressure that is still present after the discharging may lead to serious injuries.

- ▶ Check the residual pressure in the diaphragm-type accumulator after heating it up to the ambient temperature, before opening the gas filling screw.

DANGER!



Risk of injuries when disassembling under pressure!

Damage of the diaphragm-type accumulators or risk of injury in case of disassembly of the diaphragm-type accumulator, when the hydraulic system is under pressure.

- ▶ Prior to commencing any works on the diaphragm-type accumulator, depressurize the system.

DANGER!



Risk of injury and suffocation!

- ▶ Risk of injury due to uncontrolled release of a greater volume of gas and risk of injury due to entrained components.
- ▶ For checking the pre-filling pressure, always use a suitable charging and testing device.

Disposal**Disassembly of the diaphragm-type accumulator**

1. Make sure that the hydraulic system is depressurized.
2. Remove the protective cap from the gas side of the diaphragm-type accumulator.
3. Screw the prefill and test valve for hydraulic accumulators onto the gas port of the diaphragm-type accumulator. When doing this, always observe the operating instructions of the charging and testing device.
4. Discharge the pre-filling pressure on the gas side. When doing this, always observe the operating instructions of the charging and testing device.
5. The pre-filling pressure depends on the temperature. During the discharging, the diaphragm-type accumulator cools down. Wait until the accumulator heats up to ambient temperature again.
6. Discharge the pre-filling pressure generated during the heating up on the gas side. When doing this, always observe the operating instructions of the charging and testing device.
7. Remove the prefill and test valve for hydraulic accumulators from the gas port.
8. Remove the gas filling screw if required.
9. Screw the sealing cap and the protective cap back on the accumulator.
10. Fasten diaphragm-type accumulators with a capacity of 2.0 l and more on the lifting device.
11. Disconnect the diaphragm-type accumulator from the system.
12. Loosen the holding devices of the diaphragm-type accumulator.

The diaphragm-type accumulator is now disassembled.

11 Disposal

Avoid a reuse of the diaphragm-type accumulator as pressure vessel by sawing it open or drilling through it at the welded seam.

For the disposal of the diaphragm-type accumulator, the following instructions have to be complied with:

1. Remove the gas filling screw.
2. Discharge the hydraulic fluid completely.

11.1 Environmental protection

Careless disposal of diaphragm-type accumulator and of the residual fluid contained therein can cause environmental pollution.

- Dispose of the diaphragm-type accumulator in accordance with the provisions applicable in your country.
- Dispose of fluid residues according to the respective safety data sheets valid for these hydraulic fluids.

12 Extension and conversion

A conversion or extension of the diaphragm-type accumulator is not admissible. If any mechanical works are carried out on the diaphragm-type accumulator, it must no longer be operated.

13 Troubleshooting

13.1 How to proceed for troubleshooting

Always work systematically and targeted, even when under time pressure. Random, thoughtless disassembly and changing of settings can, in the worst-case-scenario, result in the inability to determine the original error cause.

First, get a general idea of how the diaphragm-type accumulator works in conjunction with the overall system.

Try to establish whether the diaphragm-type accumulator worked properly in conjunction with the overall system before the problem first occurred.

Try to determine any changes in the overall system in which the diaphragm-type accumulator is integrated:

- Were there any changes to the diaphragm-type accumulator's operating conditions or operating range?
- Have modifications (e. g. refittings) or repairs been carried out at the overall system (machine/system, electrical systems, control) or at the diaphragm-type accumulator? If so: What were they?
- Was the diaphragm-type accumulator or machine used as intended?
- How did the fault become apparent?

Try to get a clear idea of the cause of the fault. If necessary, ask the direct operator or the machine operator.

If you could not remedy the fault that occurred, please contact one of the addresses you find at <http://www.boschrexroth.com> or:

Bosch Rexroth AG

Service Industriehydraulik [Industrial hydraulics]

Competence Center Lohr

Bgm.-Dr.-Nebel-Str. 8

97816 Lohr am Main

Phone: +49 (0) 93 52 - 18 0

Troubleshooting

13.2 Fault table

Table 6: Fault table for the HAD..1X/-2X diaphragm-type accumulator

Fault	Possible causes	Remedy
Initial gas tension changes during operation	Leakage / gas port leaks	Visual examination; leak test with leak spray during final assembly
	Gas port broken/damaged	Replace the accumulator Always use the charging and testing device according to data sheet 50150 for the charging and discharging of the diaphragm-type accumulator
	Gas port function impaired due to contamination	Screw the cover caps back on after using the charging and testing device
	System-related change of pressure depending on the temperature	Temperature-dependant changes of the gas charge pressure cannot be avoided; the gas charge pressure must be chosen in accordance with the operating temperatures expected
	Reduction due to permeation	Refill gas
Cracks on the vessel	Application out of specification	Immediately stop the system; replace the accumulator: The accumulator must not be put into operation again, see chapter "11 Disposal";
	Operation outside the prescribed temperature range	Immediately stop the system; replace the accumulator: The accumulator must not be put into operation again, see chapter "11 Disposal"; check the operating temperature after commissioning
Internal corrosion	Use of inappropriate hydraulic fluids	Immediately stop the system; replace the accumulator: The accumulator must not be put into operation again, see chapter "11 Disposal"; use an appropriate hydraulic fluid
	Residual humidity in the vessel due to inappropriate storage conditions	Immediately stop the system; replace the accumulator: The accumulator must not be put into operation again, see chapter "11 Disposal"; observe the prescribed storage conditions
Oil losses at interfaces to the outside	Incorrect order of assembly in service case	Service may only be carried out by authorized specialists
	O-ring of the sealing is damaged and/or hardened due to too high oil temperature	Replace the O-ring; check the oil temperature
	Oil port cracked due to non-tension-free assembly of the diaphragm-type accumulator	Assembly without tension stress
	Inadmissible environmental force effect	Force-free assembly
	Damaging of the oil valve during transport	Visual examination after transport
Impossible to assemble	Wrong thread	Replace the parts concerned
	Thread damaged	

14 Technical data

Table 7: Technical data

Nominal value I	0.075	0.16	0.35	0.5	0.7	1.0	1.4	2.0	2.8	3.5
Maximum operating pressure	After punch									
Possible max. operating pressure (bar)					100				70	
			210	160	180		140	100		
	250	250	207	207	250	200	250	250	250	250
				250	350		350	350	350	350
Max. adm. pressure fluctuation range (bar)					93				50	
			90	90	93		80	65		
	150	120	120	120	140	115	140	140	130	130
				100	130		130	130	130	130
Ambient temperature range	-15 to +65 °C									
Max. pre-filling pressure	130 bar									

Additional technical data of the HAD..1X/-2X diaphragm-type accumulator are provided in the technical data sheet 50150.

15 Appendix

15.1 Address directory

The addresses of our sales and service network are provided under www.boschrexroth.com/adressen

15.2 Accessories

A list of accessory parts for the HAD..1X/-2X diaphragm-type accumulator is provided in the technical data sheet 50150.

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